

8. Insertion Sort with Duplicate Elements

Aim

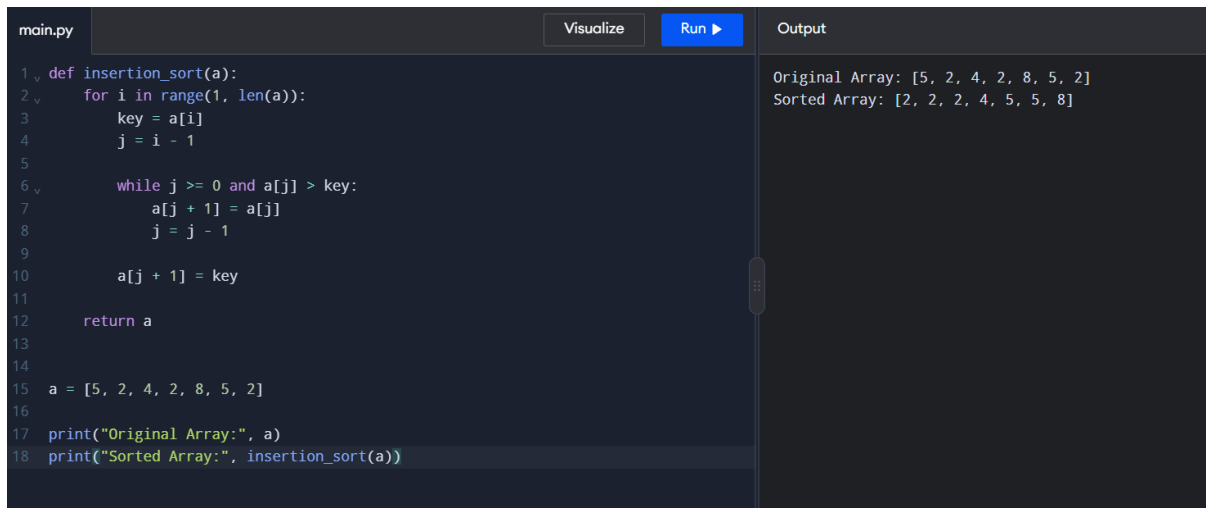
To implement Insertion Sort on an array containing duplicate elements and analyze its time and space complexity.

Algorithm

1. Start from the second element.
2. Store the current element as key.
3. Compare the key with elements before it.
4. Shift larger elements one position to the right.
5. Insert the key into its correct position.
6. Repeat until all elements are sorted.

Procedure

1. Read an array containing duplicate elements.
2. Select the second element as the key.
3. Compare it with the sorted portion.
4. Shift larger elements to the right.
5. Insert the key at the correct position.
6. Repeat for all remaining elements.
7. Display the sorted array.



```
main.py Visualize Run ▶ Output
1 def insertion_sort(a):
2     for i in range(1, len(a)):
3         key = a[i]
4         j = i - 1
5
6         while j >= 0 and a[j] > key:
7             a[j + 1] = a[j]
8             j = j - 1
9
10        a[j + 1] = key
11
12    return a
13
14
15 a = [5, 2, 4, 2, 8, 5, 2]
16
17 print("Original Array:", a)
18 print("Sorted Array:", insertion_sort(a))
```

Original Array: [5, 2, 4, 2, 8, 5, 2]
Sorted Array: [2, 2, 2, 4, 5, 5, 8]

Result

Thus, the given array containing duplicate elements was successfully sorted using Insertion Sort, and its time complexity was analyzed.